

SAT Unit 2.1.1

Exponents & Radicals

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Overview

Definition of Exponents

Definition: An exponential expression has the form x^n , where:

- x is the **base**
- n is the **exponent** or power

This represents the base x being multiplied by itself n times.

$$x^n = \underbrace{x \cdot x \cdot x \cdots x}_{n \text{ times}}$$

Laws of Exponents (Part 1)

Basic Properties

- **Identity:** $x^1 = x$ (e.g., $3^1 = 3$)
- **Zero Exponent:** $x^0 = 1$ (e.g., $3^0 = 1$)

Multiplication and Division Rules

- **Product Rule:** $x^m \cdot x^n = x^{m+n}$ (e.g., $3^4 \cdot 3^5 = 3^9$)
- **Quotient Rule:** $\frac{x^m}{x^n} = x^{m-n}$ (e.g., $\frac{3^7}{3^3} = 3^4$)

Laws of Exponents (Part 2)

Power Rules

- **Power of a Power:** $(x^m)^n = x^{mn}$ (e.g., $(3^2)^4 = 3^8$)
- **Power of a Product:** $(xy)^m = x^m y^m$ (e.g., $(2 \cdot 3)^3 = 2^3 \cdot 3^3$)
- **Power of a Quotient:** $\left(\frac{x}{y}\right)^m = \frac{x^m}{y^m}$ (e.g., $\left(\frac{2}{3}\right)^3 = \frac{2^3}{3^3}$)

Negative Exponent

- **Rule:** $x^{-m} = \frac{1}{x^m}$ (e.g., $3^{-4} = \frac{1}{3^4}$)

Question: Product of Exponential Expressions

Which expression represents the product of

$$(b^6c^{-2}d^{-5}) \quad \text{and} \quad (b^8c^{-3} + c^4d^5),$$

where b , c , and d are positive?

Options:

- A. $b^{14}c^{-5} + c^2d^{-10}$
- B. $b^{14}c^{-5} + c^2$
- C. $b^{14}c^{-5}d^{-5} + b^6c^2$
- D. $b^{14}c^{-5}d^{-5} + c^2$

Answer and Explanation

Explanation:

We are asked to find the product of:

$$(b^6c^{-2}d^{-5}) \quad \text{and} \quad (b^8c^{-3} + c^4d^5)$$

Distribute the first term across both terms in the second expression:

$$b^6c^{-2}d^{-5} \cdot b^8c^{-3} = b^{6+8}c^{-2-3}d^{-5} = b^{14}c^{-5}d^{-5}$$

$$b^6c^{-2}d^{-5} \cdot c^4d^5 = b^6c^2d^0 = b^6c^2$$

Putting it together:

$$b^{14}c^{-5}d^{-5} + b^6c^2$$

Now check the answer choices. Only choice **C** has this exact expression.

Final Answer: C. $b^{14}c^{-5}d^{-5} + b^6c^2$



Radicals and Roots

Definition: A radical is the inverse of an exponent.

$$\sqrt[n]{x} = x^{\frac{1}{n}}$$

- $\sqrt{x} = x^{\frac{1}{2}}$ (square root)
- $\sqrt[3]{x} = x^{\frac{1}{3}}$ (cube root)
- $\sqrt[4]{x^3} = x^{\frac{3}{4}}$ (fractional exponent)

Properties of Radicals

- $\sqrt{a} \cdot \sqrt{b} = \sqrt{ab}$
- $\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$
- $\sqrt[n]{x^m} = x^{\frac{m}{n}}$

Important: All radicals can be rewritten using fractional exponents!

Rationalizing Denominators

Goal: Remove square roots from denominators.

$$\frac{1}{\sqrt{2}} \Rightarrow \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

Use the **conjugate** if the denominator is a binomial:

$$\frac{1}{\sqrt{2} + 1} \cdot \frac{\sqrt{2} - 1}{\sqrt{2} - 1} = \frac{\sqrt{2} - 1}{(\sqrt{2})^2 - 1^2} = \frac{\sqrt{2} - 1}{1}$$

Question

Simplify the expression:

$$\left(\frac{27x^6y^3}{81x^2y^9}\right)^{\frac{1}{3}}$$

Solution

Solution:

$$= \left(\frac{27}{81} \cdot x^{6-2} \cdot y^{3-9} \right)^{\frac{1}{3}} = \left(\frac{1}{3} \cdot x^4 \cdot y^{-6} \right)^{\frac{1}{3}} = \frac{x^{\frac{4}{3}}}{3 \cdot y^2}$$

Question: Rational Exponents and Roots

If

$$\frac{\sqrt{x^5}}{\sqrt[3]{x^4}} = x^{\frac{a}{b}}$$

for all positive values of x , what is the value of $\frac{a}{b}$?

Solution

Solution:

We use the property of radicals and exponents:

$$\sqrt{x^5} = x^{5/2}, \quad \sqrt[3]{x^4} = x^{4/3}$$

$$\frac{x^{5/2}}{x^{4/3}} = x^{5/2-4/3}$$

Find a common denominator:

$$\frac{5}{2} = \frac{15}{6}, \quad \frac{4}{3} = \frac{8}{6}$$

$$x^{15/6-8/6} = x^{7/6}$$

Thus,

$$\frac{a}{b} = \frac{7}{6}$$

Answer:

$$\boxed{\frac{7}{6}}$$

Question: Exponent and Radical Expression Equivalence

For what value of x is the given expression equivalent to $(70n)^{30x}$, where $n > 1$?

$$\sqrt[5]{70n \left(\sqrt[6]{70n} \right)^2}$$

Solution: Simplifying the Expression

We begin with:

$$\sqrt[5]{70n \left(\sqrt[6]{70n} \right)^2} = \sqrt[5]{70n \cdot (70n)^{2/6}} = \sqrt[5]{(70n)^{1 + \frac{2}{6}}} = \sqrt[5]{(70n)^{\frac{4}{3}}}$$

Now apply the outer root:

$$\left((70n)^{\frac{4}{3}} \right)^{1/5} = (70n)^{\frac{4}{15}}$$

We are told this is equal to:

$$(70n)^{30x} \Rightarrow 30x = \frac{4}{15} \Rightarrow x = \frac{4}{450} = \frac{2}{225}$$

Answer:

$$\boxed{\frac{2}{225}}$$



Question

If $(5^3)^{4k} = \left(5^{\frac{1}{3}}\right)^{24}$, what is the value of k ?

Answer

We simplify both sides:

$$(5^3)^{4k} = 5^{12k}, \quad \left(5^{\frac{1}{3}}\right)^{24} = 5^{\frac{24}{3}} = 5^8$$

So,

$$5^{12k} = 5^8 \Rightarrow 12k = 8 \Rightarrow k = \frac{2}{3}$$

Correct answer: B

Question

Which expression is equivalent to 4^{m+2} ?

- A) 16^m
- B) $16 + 4^m$
- C) $8(4^m)$
- D) $16(4^m)$

Answer

We simplify the original expression:

$$4^{m+2} = 4^m \cdot 4^2 = 4^m \cdot 16 = 16(4^m)$$

Correct answer: D

Question

$$\sqrt[8]{41k} \left(\sqrt[9]{41k} \right)^{12}$$

For what value of x is the given expression equivalent to $(41k)^{25x}$, where $k > 1$?

Answer

Correct answer: $x = \frac{7}{120}$

Question

$$\frac{10\sqrt[6]{9^3 \cdot x^{48}}}{\sqrt[4]{6^4 x}}$$

The given expression is equivalent to ax^b , where $a > 0$, $b > 0$, and $x > 1$. What is the value of ab ?

Answer

We simplify numerator and denominator:

Numerator:

$$10 \cdot \sqrt[6]{9^3 \cdot x^{48}} = 10 \cdot (9^3 \cdot x^{48})^{1/6} = 10 \cdot 9^{1/2} \cdot x^8$$

Since $9 = 3^2$, so $9^{1/2} = 3$, hence numerator becomes:

$$10 \cdot 3 \cdot x^8 = 30x^8$$

Denominator:

$$\sqrt[4]{6^4 x} = (6^4 x)^{1/4} = 6 \cdot x^{1/4}$$

Entire expression:

$$\frac{30x^8}{6x^{1/4}} = 5x^{8-1/4} = 5x^{\frac{31}{4}}$$

So $a = 5$, $b = \frac{31}{4}$

$$ab = 5 \cdot \frac{31}{4} = \frac{155}{4}$$

Question

$$\sqrt[4]{x^3} \cdot \sqrt[12]{x^5}$$

The given expression is equivalent to $\sqrt[n]{x^m}$, where m and n are positive constants each less than 10 and $x > 1$. What is the value of $\frac{m}{n}$?

Answer

We simplify the expression:

$$\sqrt[4]{x^3} \cdot \sqrt[12]{x^5} = x^{\frac{3}{4}} \cdot x^{\frac{5}{12}} = x^{\frac{3}{4} + \frac{5}{12}}$$

To add the exponents:

$$\frac{3}{4} = \frac{9}{12}, \quad \frac{9}{12} + \frac{5}{12} = \frac{14}{12} = \frac{7}{6}$$

So the expression becomes:

$$x^{\frac{7}{6}} = \sqrt[6]{x^7}$$

Therefore, $m = 7$, $n = 6$, and

$$\frac{m}{n} = \frac{7}{6}$$

Correct answer: $\frac{7}{6}$

Thank You!

We appreciate your time and focus.

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